Proceedings of the ASME 2016 International Mechanical Engineering Congress and Exposition
IMECE2016
November 11-17, 2016, Phoenix, Arizona, USA

IMECE2016-66719

A VIBRATION COMPENSATION METHOD FOR ABSOLUTE GRAVIMETERS

Guan Wang
Tsinghua University
Beijing 100084, China

Hua Hu
Tsinghua University
Beijing 100084, China

Kang Wu
Tsinghua University
Beijing 100084, China

Lijun Wang
Tsinghua University
Beijing 100084, China

ABSTRACT

The laser interferometer is used to track the falling object in a freefall absolute gravimeter, which could be disturbed by the vibration from the ground. Thus, the vibration compensation method is often used to reduce the influence of the vibration. Typically, a sensor (broadband seismometer) is used to record the vibration. But the measured 'vibration' $N_m(t)$ does not equal the motion of the reference corner cube $N(t)$. Because there exists a transfer function $G(s)$ making $N_m(s) = G(s)N(s)$. Traditionally, $G(s)$ is assumed to be equal to the transfer function of the sensor, which can be achieved with the help of other equipment. But the assumption is not reasonable and the process of calculating the transfer function is complicated. A novel vibration compensation method without any other equipment is proposed in this paper. In this method, $G(s)$ is simplified to estimate $N(t)$ using $N'(t) = AN_m(t - \tau)$, which is used for compensation. The gain A and delay τ can be obtained by analysis of the data acquired by the absolute gravimeter. The experiments are conducted with the homemade absolute gravimeter T-1 and repeated for 75 times. The standard deviation of the uncompensated results is 3276 μGal (1 μGal = 1×10^{-8} m/s^2), while that of the compensated results is 167 μGal. The compensation method not only achieves a reduction by nearly a factor of 20, but also can be simply used without any other equipment. The results indicate that the method basically meets the demands of absolute gravimeters. In the future, it may be applied to dynamic absolute gravity measurement and take the place of vibration isolators.

Keywords Vibration compensation; absolute gravimeter; seismometer; transfer function; laser interferometry

INTRODUCTION

Absolute gravimetry is the measurement of gravitational acceleration (g, about 9.8 m/s^2), which is an important method to obtain high precision Earth gravity field [1]. Absolute gravimeter is a kind of precision instruments to measure the gravitational acceleration [2]. Since the 20th century, with the development of laser and atomic frequency standard, the research of absolute gravimeters has developed rapidly [3]. Nowadays, the high precision absolute gravimeter plays an important role in many fields such as geodetics, geophysics, geodynamics, geological exploration, navigation, metrology, *etc.* [4,5]. For a classical free-fall absolute gravimeter, the trajectory of the free-falling object is measured with a laser interferometer and an atomic clock [6]. And the gravitational acceleration is obtained by quadratically fitting the time and distance pairs [7]. Theoretically, laser interferometry needs an inertial reference point, as the measurement will be inaccurate if the reference point is moving [8]. But the reference point, which is a corner cube for absolute gravimeters, is always greatly disturbed by the vibration from the ground [9]. Consequently, it is necessary for high precision absolute gravimeters to reduce the influence of the vibration.

Vibration from the ground can be divided into three parts according to the frequency. Above 0.5 Hz, the main source of vibration is from the surroundings like human activity and the swing of buildings. From 0.05 Hz to 0.5 Hz, the main source of vibration is the surface and body waves in the earth caused by ocean waves. Below 0.05 Hz, the main source of vibration is atmospheric pressure fluctuations. The resonance frequency of most traditional vibration isolators is about 1 Hz, which is not enough to isolate the vibration above [10-13]. Thus, there are two ways to reduce the influence of vibration for absolute gravimeters. One is to develop an ultra-low frequency vibration isolator, whose resonance frequency should be lower than 0.03 Hz [14, 15]. The other is to compensate the results by recording the vibration with a broadband sensor [16].

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use

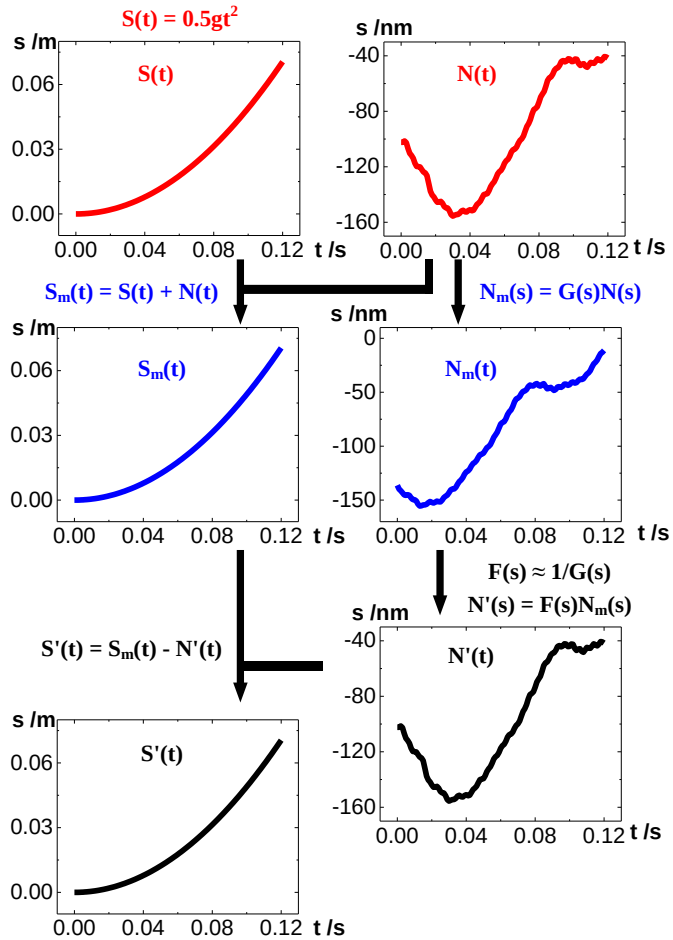


Figure 1. Principle of the vibration compensation.

The ultra-low frequency vibration isolator is widely used with its good performance [17]. For example, the FG5/FG5X gravimeters made by Microg-Lacoste in America use a vibration isolator called 'Super Spring'. Other research organizations such as Stanford University and Wuhan Institute of Physics and Mathematics in China, develop vibration isolators for their own absolute gravimeters as well [18-20]. The disadvantage of ultra-low frequency vibration isolator is that it is complicated and sensitive to disturbance and can only be used on the ground or stable platform. There are also some absolute gravimeters using vibration compensation method with a broadband seismometer, such as NIM-3 made by Chinese National Institute of Metrology, the atomic gravimeters made by Zhejiang University of Technology in China and the AQG made by Muquans in France [21-23]. Although the effect of compensation method is not as good as that of isolators, it could greatly simplify the system of absolute gravimeter and make it smaller and applicable to more occasions.

The principle of compensation method is simple, as shown in Fig. 1. $S(t) = 0.5gt^2$, is the actual trajectory of the free-falling object, which can be described with a quadratic function. $N(t)$ is the motion of reference corner cube caused by the vibration which impacts the laser interferometry. So the measured trajectory is $S_m(t) = S(t) + N(t)$. Meanwhile, a sensor is recording the vibration from the ground $N_m(t)$. But $N_m(t)$ is not $N(t)$, as there exists a transfer function $G(s)$ making $N_m(s) = G(s)N(s)$. Thus, we need to calculate $G(s)$ or find $F(s)$ making $F(s) \approx 1/G(s)$ to compute $N'(t)$. Finally, $S(t)$ is estimated with $S'(t) = S_m(t) - N'(t)$. The essence of the method is to calculate $G(s)$ or $F(s)$. The traditional way is to regard $G(s)$ as the transfer function of the sensor and calculate it with the sine scanning excited method. The simplification is inaccurate as $G(s)$ is also determined by reference corner cube and their positions. In addition, the process of computing sensor's transfer function is difficult without a precision ultra-low frequency vibration source.

A novel vibration compensation method is proposed in this paper that $G(s)$ is simplified and easily calculated. The paper is organized as follows. The principle of the method and the solving process is detailed described in section 2. Section 3 shows the experiments and results. Finally, we give the conclusions in section 4.

PRINCIPLE

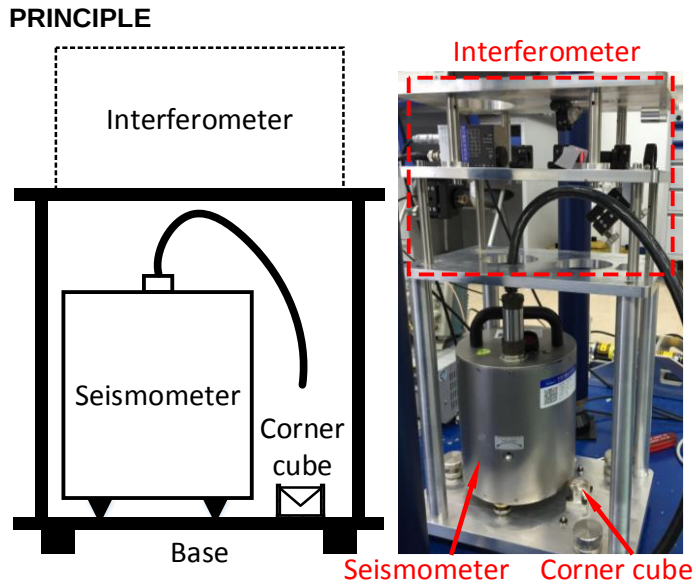


Figure 2. The positions of seismometer and reference corner cube.

Seismometer

The seismometer is the most important component for the vibration compensation method. Its performance affects the results seriously. The seismometer used is CMG-3ESP made by Guralp Systems Ltd., which could be sensitive to ground vibrations in the frequency range 0.016-50 Hz with 2×750 V/m/s velocity sensitivity. The output analog velocity signal of the seismometer is collected by a high-accuracy Analog to Digital Converter (ADC) and then integrated to displacement signal. There is no need to calibrate the output velocity signal, as the deviation does not influence the quadratic coefficient of the displacement signal. The seismometer is placed on a base, where the reference corner cube is fastened, as shown in Fig. 2. The relative position of the seismometer and the corner cube is fixed to guarantee the transfer function $G(s)$ constant.

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use

Transfer function

As $G(s)$ is complicated and hard to calculate, it could be simplified to a time delay system. Thus we assume $N(t) \approx N'(t) = AN_m(t-\tau)$, in which A is the gain and τ is the delay. The reason of the assumption is that vibration mainly distributes in 0.1-10 Hz, as shown in Fig. 3. Then, the seismometer can be regarded as a bandpass filter, whose cut-off frequencies are 0.016 and 50 Hz. So the amplitude frequency characteristic and the delay of the seismometer from 0.1 Hz to 10 Hz are both approximately constant.

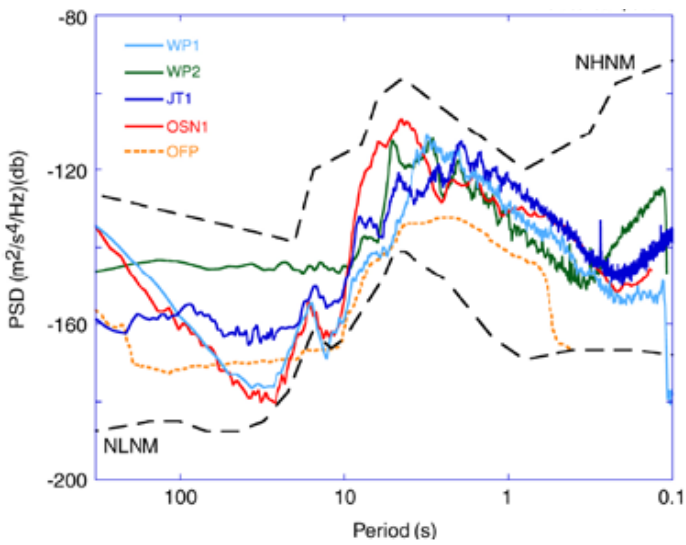


Figure 3. Vertical components of seismic noises. (PSD: Power Spectral Density; NHNM: New High Noise Model; NLNM: New Low Noise Model; WP1, WP2, JT1, OSN-1, OFP are short for different observatories.) [24].

Delay τ

When the object is free falling in the vacuum chamber, the motion of the object $S(t)$ can be described as a quadratic function theoretically. The motion of the reference corner cube caused by the vibration is $N(t)$. Thus, the signal measured by the interferometer is $S_m(t) = S(t) + N(t)$, as shown in Fig. 5(a). The gravitational acceleration g_m and the residuals $R_m(t)$ can be achieved by quadratically fitting $S_m(t)$. Similarly, the residuals $R_N(t)$ and the quadratic coefficient a are achieved by quadratically fitting $N(t)$. We find out that $R_m(t) = R_N(t)$ and $g_m = g + 2a$. Therefore, the delay can be computed according to the relation between residuals and vibration.

In T-1 absolute gravimeter, a trigger signal is generated when the velocity of the free falling object reaches the threshold. When triggered, the ADC begins to collect the output signal of the interferometer. The collection lasts for 120 ms. Meanwhile, the signal from the seismometer begins to be collected 40 ms before the trigger signal and lasts for 200 ms. Thus, $S_m(t)$ is 120 ms long and $N_m(t)$ is 200 ms long. The process of calculation is shown in Fig. 5(b). A 120 ms long window moves on $N_m(t)$ with a step of 0.01 ms. For each step, there is a $N_i(t)$ and by quadratically fitting $N_i(t)$, we can achieve its residuals $R_i(t)$. Then the correlation coefficient between $R_i(t)$ and $R_m(t)$ is calculated and the maximum can be obtained. Based on the maximum, we can calculate the τ and get the corresponding vibration $N_\tau(t)$ of $S_m(t)$, which will be used in the following for the compensation. By this way, τ can be solved with one drop theoretically. But in practice, we could solve τ_i for each drop of a set and use their mean value as the τ to improve the accuracy.

Gain *A*

Vibration from the ground is random noise, which affects the standard deviation of the results of an absolute gravimeter. In theory, better compensation method leads to smaller standard deviation. Thus, we use T-1 absolute gravimeter to carry out multiple drops as a set, calculate the delay τ and try different gains A_i to compensate. For each A_i we can calculate the standard deviation σ_i of the set, and the A with minimal σ is what we want.

Input: 25 drops A_0, d. | $S_{m1}(t)$ $N_{\tau 1}(t)$ | ..... | $S_{mj}(t)$ $N_{\tau j}(t)$ | ..... | $S_{m25}(t)$ $N_{\tau 25}(t)$

Step 1: $A_k = A_0 + kd$, $(k = -20\sim20)$

Step 2: A_{-20} : ..... $S'_{-20j}(t) = S_{mj}(t) - A_{-20}N_{\tau j}(t)$ .....

⋮

A_k : ..... $S'_{kj}(t) = S_{mj}(t) - A_kN_{\tau j}(t)$ .....

⋮

A_{20} : ..... $S'_{20j}(t) = S_{mj}(t) - A_{20}N_{\tau j}(t)$ .....

Step 3: A_{-20} : ..... g'_{-20j} ..... σ_{-20}

⋮

A_k : ..... g'_{kj} ..... σ_k

⋮

A_{20} : ..... g'_{20j} ..... σ_{20}

Step 4: Find minimum σ and A^*.
If $d > 0.01A^*$, define $A_0 = A^*$, $d = d/20$, go to Step 1.

Output: A^*

Figure 4. The process of solving A.

The process of computing A is shown in Fig. 4. Firstly, we carry out 25 drops as a set, record $S_{mj}(t)$ and find out $N_{\tau j}(t)$ (j = 1~25). Then we set an initial value A_0 according to the parameter of the seismometer and choose a proper d as the common difference. The process is as follows.

Step 1: Generate an arithmetic progression $A_k = A_0 + kd$, (k = -20~20).

Step 2: For each A_k, compute $S_{jk}'(t) = S_{mj}(t) - A_kN_{\tau j}(t)$.

Step 3: Achieve g_{jk}' by quadratically fitting $S_{jk}'(t)$ and calculate the standard deviation of g_{jk}' (j = 1~25) as σ_k.

Step 4: Find A^* with minimal σ.

At this point, make $A_0 = A^*$, $d = d/20$, and cycle the procedures above until $d < 0.01A^*$. Finally, A^* is the gain.

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use

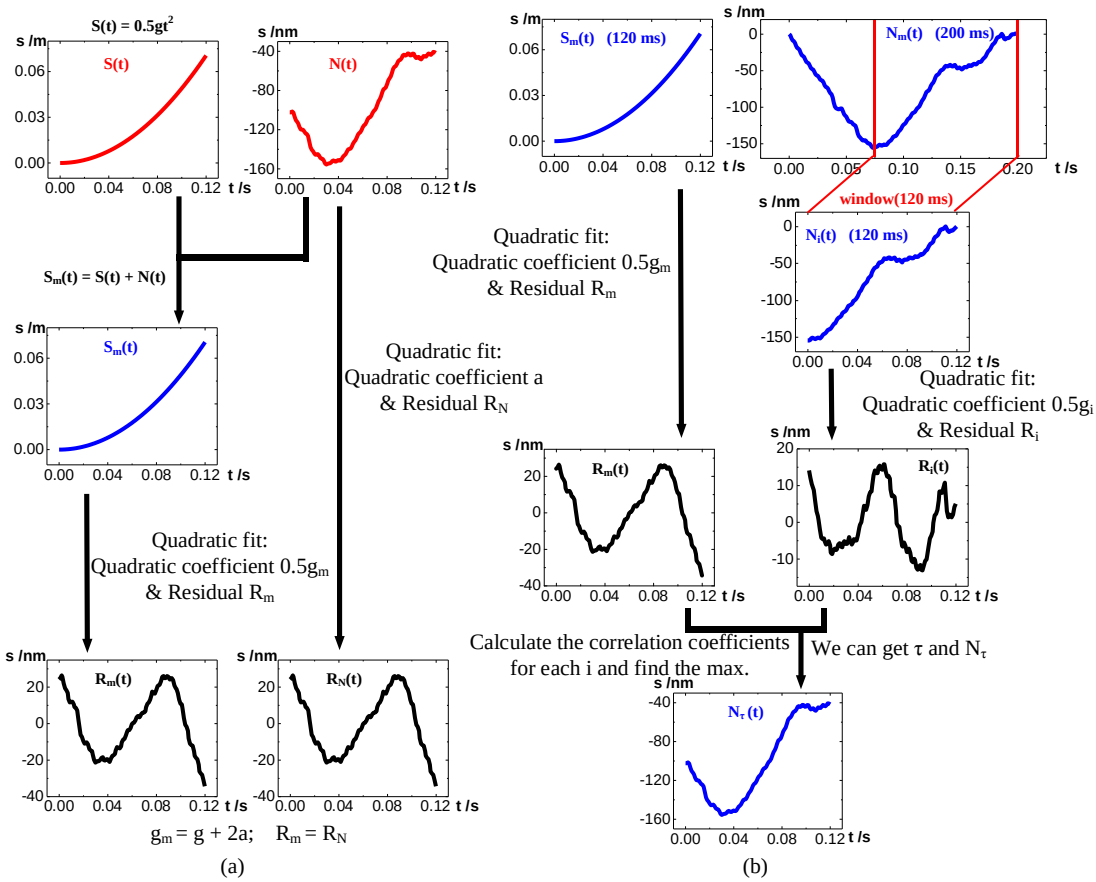


Figure 5. (a) Relation between residuals and vibration. (b) The process of solving delay.

EXPERIMENTAL RESULTS

To evaluate the performance of the proposed compensation method, it is applied to T-1 absolute gravimeter in Tsinghua University, as T-1 has been proved stable by long-time operation. The measurement results with 'Super Spring' vibration isolator in the same location are used for comparison as well. The experimental results consist of two parts: one is to evaluate the solution of gain and delay, the other is to evaluate the compensated results.

T-1 absolute gravimeter

The T-1 absolute gravimeter, as shown in Fig. 6, is a classical free-fall style prototype designed and built in Tsinghua University [25]. T-1 consists of a vacuum chamber for the free falling object, a laser interferometer to measure displacement, an ultra-low frequency vibration isolator called 'Super Spring', a high-speed ADC and a data processing software. With precision mechanical design, the combined standard uncertainty of T-1 is about 5 μGal. T-1 also participated in the European Comparison of Absolute Gravimeters 2011 (ECAG-2011) and the International Comparison of Absolute Gravimeters 2013 (ICAG-2013), whose results agreed with the reference value [26]. Thus, it is persuasive to evaluate the compensation method with T-1 absolute gravimeter.

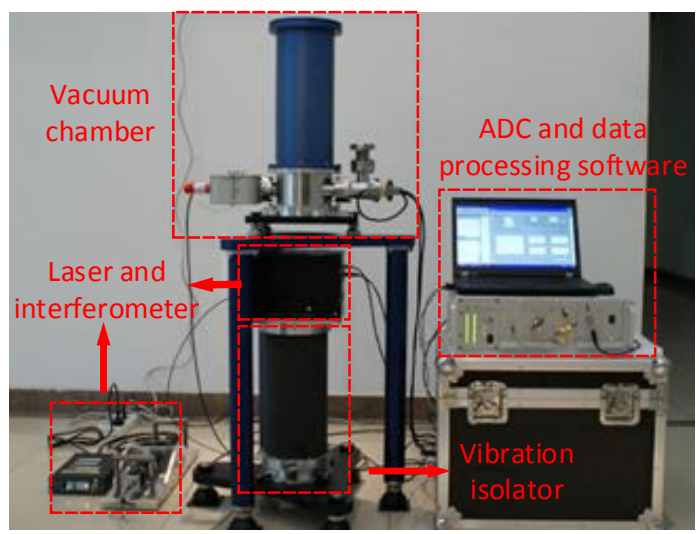


Figure 6. T-1 absolute gravimeter.

Estimation of the delay and gain

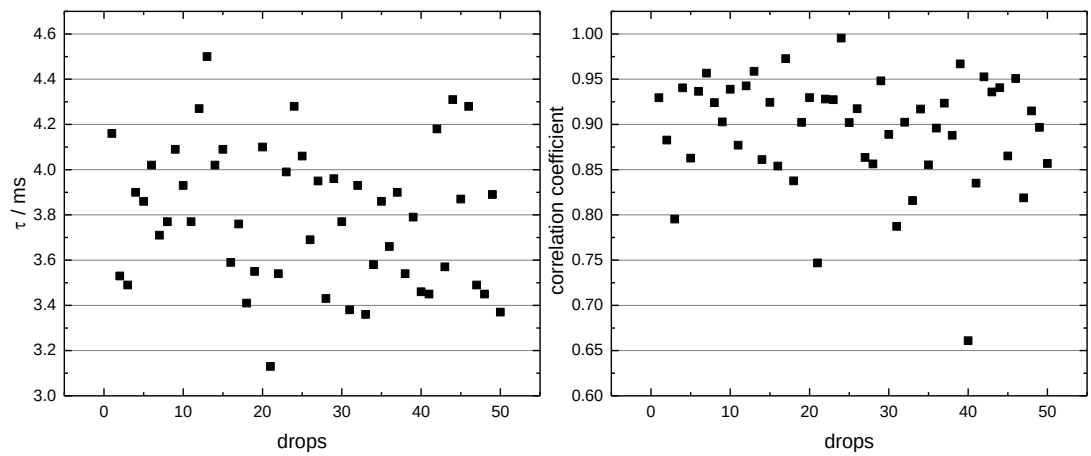
If the assumption and the solving process are right, the delay for each drop should be approximately equal. Thus, we continually drop 50 times and the delays calculated by the compensation method are shown in Fig. 7(a). The average is 3.79

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use

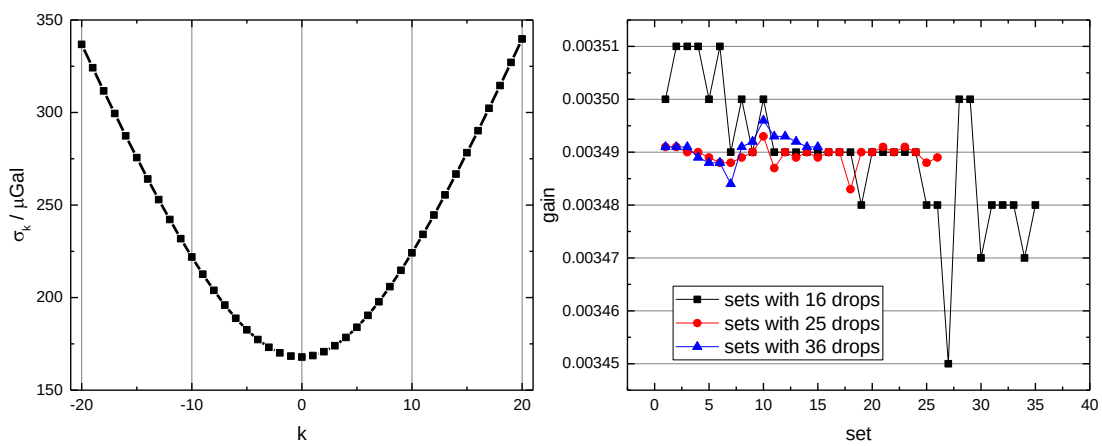
ms and the standard deviation is 0.31 ms, which is small compared to the average. In addition, the maximal correlation coefficients of each drop are shown in Fig. 7(b). The average is 0.89, which is quiet a strong correlation. Thus, the method of computing delay is verified.

As stated earlier, the gain can be calculated with a set of drops. For each set, σ_k is described as quadratic function of k approximately, as shown in Fig. 8(a), which indicates the existence of a minimal σ described in Fig. 4. Similarly, if the assumption and the solving process are correct, the gain for different sets should be approximately equal. Firstly, we calculate with the set of 25 consecutive drops so that we have 26 sets from the 50 drops. All the gains are shown in Fig. 8(b). The average is 0.0034896 and the standard deviation is 1.96×10^{-6}. Then, we calculate with the sets of 16 drops and 36 drops successively. Both of the averages are about 0.00349. But for the sets with 16 drops, the standard deviation is much larger, which means 16 drops are not enough. Finally, the method of solving gain is verified as well.



(a) (b)

Figure 7. (a) Delays calculated of 50 drops. (b) The maximum correlation coefficients of 50 drops.



(a) (b)

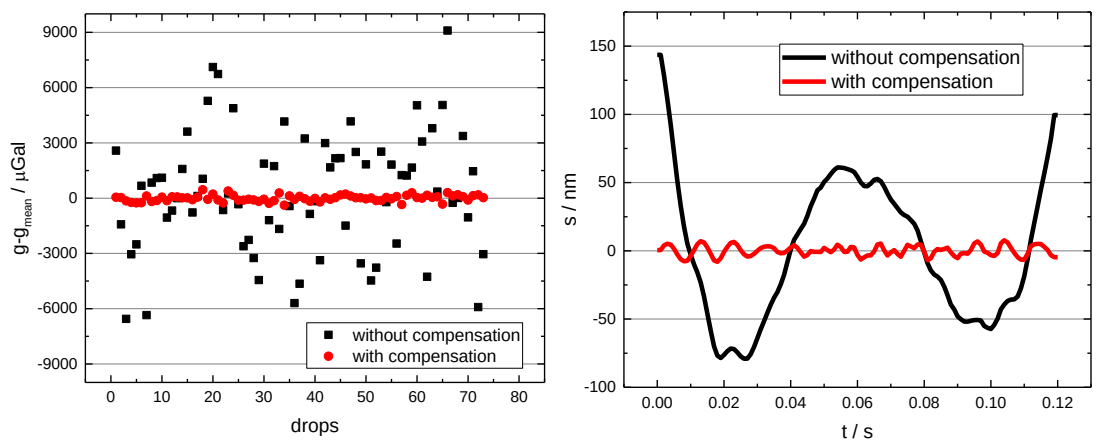
Figure 8. (a) Relation between σ_k and k. (b) Gains calculated of different sets.

Compensation results

We carry out compensation with A = 0.0035 and τ = 3.79 ms obtained above. The uncompensated and compensated results for 75 drops (2 drops are invalid and eliminated) are shown in Fig. 9(a). The standard deviation of uncompensated results is 3276 μGal, while that of compensated results is only 167 μGal. The compensation method nearly achieves a reduction by a factor of 20. The residuals are compensated as well, the amplitude of residuals without compensation is about 150 nm, while that with compensation is no more than 10 nm, as shown in Fig. 9(b).

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use



(a) (b)

Figure 9. (a) Comparison of the results with compensation and without compensation. (b) Comparison of the residuals with compensation and without compensation.

The comparison results with 'Super Spring' are shown in Fig.10, which could represent the best level of the vibration isolator at present. The 'Super Spring' indeed has better performance, but the difference is not so much. It is indicated that the proposed compensation method achieves comparative performance in comparison with the vibration isolator, while our method does not need any other auxiliary equipment. For absolute gravimeters, the compensation method could meet the demands of measurement with the precision of 10 μGal level.

CONCLUSIONS

A novel vibration compensation method is presented in this paper, including its principle and experimental results. The essence of this method is to simplify the transfer function $G(s)$ and propose a simple way to compute it. In theory, the gain and the delay can be computed with only 25 drops without any other equipment. Thus, it can still work when the measurement environment changed greatly. The superiority of this method is that it is simpler than traditional compensation methods and vibration isolators, but also has significant performance.

With the development of technology, the demand for the absolute gravimeter applied on moving platforms is increasing rapidly. Simpler and smaller is also a development direction of the absolute gravimeter. For these applications, the vibration compensation is more suitable than the vibration isolator. Thus, this vibration compensation method is hopefully widely used in the future.

ACKNOWLEDGMENTS

The authors acknowledge the financial support of Tsinghua University Initiative Scientific Research Program (Grant No. 20131089245).

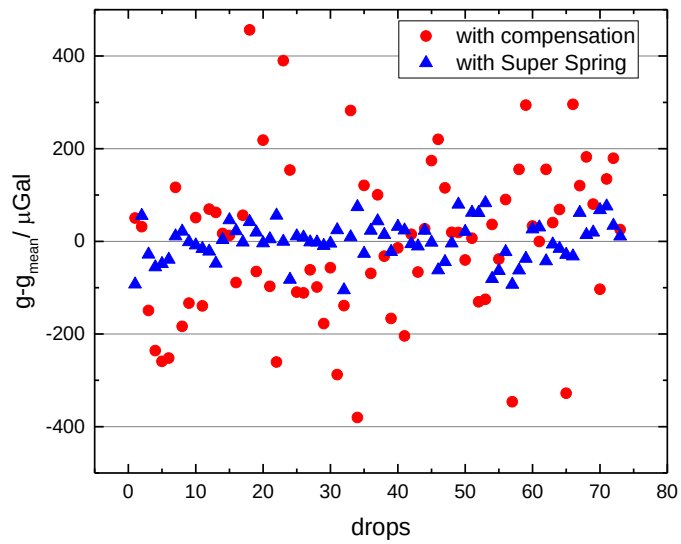


Figure 10. Comparison of the results with compensation and with 'Super Spring'.

REFERENCES

[1] Marson, I., and Faller, J. E., 1986, "g - the Acceleration of Gravity: Its Measurement and Its Importance," J. Phys. E: Sci. Instrum., **19**(1), pp. 22-32.

[2] Wang, Q. S., 2003, *Gravitology (in Chinese)*, Seismological Press, Beijing, China, Chap. 14.

[3] Faller, J. E., 2002, "Thirty Years of Progress in Absolute Gravimetry: a Scientific Capability Implemented Technological Advances," Metrologia, **39**(5), pp. 425–428.

[4] Steiner, R. L., Williams, E. R., Newell, D. B., and Liu, R., 2005, "Towards an electronic kilogram: an improved measurement of the Planck constant and electron mass," Metrologia, **42**(5), pp. 431–441.

[5] Timmen, L., Gitlein, O., Klemann, V., & Wolf, D., 2011, "Observing Gravity Change in the Fennoscandian Uplift

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use

Area with the Hanover Absolute Gravimeter," Pure Appl. Geophys., **169**(8), pp. 1331–1342.
[6] Hu, H., Wu, K., Shen, L., Li, G., and Wang, L. J., 2012, "A New High Precision Absolute Gravimeter," Acta Phys. Sin., **61**(9), pp. 544-551.
[7] Niebauer, T. M., Sasagawa, G. S., Faller, J. E., Hilt, R., and Klopping, F., 1995, "A New Generation of Absolute Gravimeters," Metrologia, **32**(3), pp. 159-180.
[8] Saulson, P. R., 1984, "Vibration isolation for broadband gravitational wave antennas," Review of Scientific Instruments, **55**(8), pp. 1315.
[9] Robertson, N. A., Drever, R. W. P., Kerr, I. and Hough, J., 1982, "Passive and active seismic isolation for gravitational radiation detectors and other instruments," Journal of Physics E: Scientific Instruments, **15**(10), pp. 1101–1105.
[10] Sorrells, G. G. and Douze, E. J., 1974, "A preliminary report on infrasonic waves as a source of long-period seismic noise," Journal of Geophysical Research, **79**(32), pp. 4908–4917.
[11] Haubrich, R. A. and McCamy, K., 1969, "Microseisms: Coastal and pelagic sources," Reviews of Geophysics, **7**(3), pp. 539.
[12] Agnew, D. C., 1986, "Strainmeters and tiltmeters," Reviews of Geophysics, **24**(3), pp. 579.
[13] Cessaro, R. K., 1994, "Sources of Primary and Secondary Microseisms," Bull. Seismol. Soc. Am., **84**(1), pp. 142-148.
[14] Nelson, P. G., 1991, "An active vibration isolation system for inertial reference and precision measurement," Review of Scientific Instruments, **62**(9), pp. 2069.
[15] Winterflood, J., Blair, D. and Slagmolen, B, 2002, "High performance vibration isolation using springs in Euler column buckling mode," Physics Letters A, **300**(2-3), pp. 122–130.
[16] Brown, J. M., Niebauer, T. M. and Klingele, E, 2001, "Towards a Dynamic Absolute Gravity System," International Association of Geodesy Symposia, 223–228.
[17] Newell, D. B., Richman, S. J., Nelson, P. G., Stebbins, R. T., Bender, P. L., Faller, J. E. and Mason, J., 1997, "An ultra-low-noise, low-frequency, six degrees of freedom active vibration isolator," Review of Scientific Instruments, **68**(8), pp. 3211.
[18] Hensley, J. M., Peters, A. and Chu, S., 1999, "Active low frequency vertical vibration isolation," Review of Scientific Instruments, **70**(6), pp. 2735.
[19] Rinker, R. and Faller, J., 1981, "'Super Spring'-A Long Period Vibration Isolator," Proceedings of Precision Measurement and Fundamental Constants Gaithersburg, the USA, June 8-12, 1981 p411
[20] Tang, B., Zhou, L., Xiong, Z., Wang, J. and Zhan, M., 2014, "A programmable broadband low frequency active vibration isolation system for atom interferometry," Review of Scientific Instruments, **85**(9), pp. 093109.
[21] Wu, S., Li, C., Su, D., Feng, J. and Xu, J., 2015, "Measurement time interval based on FPGA in NIM-3 absolute gravimeter," Ninth International Symposium on Precision Engineering Measurement and Instrumentation.
[22] Wu, B., Wang, Z., Cheng, B., Wang, Q., Xu, A. and Lin, Q., 2014, "The investigation of a μGal-level cold atom gravimeter for field applications," Metrologia, **51**(5), pp. 452–458.
[23] Gillot, P., Francis, O., Landragin, A., Pereira Dos Santos, F. and Merlet, S., 2014, "Stability comparison of two absolute gravimeters: optical versus atomic interferometers," Metrologia, **51**(5), pp. L15.
[24] Bormann, P. and Wielandt, E., 2013, "Seismic Signals and Noise," New Manual of Seismological Observatory Practice 2 (NMSOP2), GFZ, pp. 1-62.
[25] Li, G., Hu, H., Wu, K., Wang, G. and Wang, L. J., 2014, "Ultra-low frequency vertical vibration isolator based on LaCoste spring linkage," Review of Scientific Instruments, **85**(10), pp. 104502.
[26] Li, Z., Hu, H., Wu, K., Li, G., Wang, G. and Wang, L. J., 2015, "Self-attraction effect and correction on the T-1 absolute gravimeter," Metrologia, **52**(6), pp. 793–799.

Copyright © 2016 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 08/14/2017 Terms of Use: http://www.asme.org/about-asme/terms-of-use